



Department of Mathematics, Physics, and Computer Science
College of Science and Mathematics

UNIVERSITY OF THE PHILIPPINES MINDANAO

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Degree Program BS Computer Science
Date Effective 1 June 2018
Approved by UP Board of Regent

COURSE SYLLABUS IN CMSC 3 – INTRODUCTION TO WEB DESIGN

UP Statement of the Philosophy of Education and Graduate Attributes <i>Approved by the Board of Regents on 28 November 2019</i> A UP education seeks to produce graduates imbued with an abiding sense of responsibility to their people and nation, the skills and mindsets to improve human life, and a commitment to the freedom and welfare of all. Aside from mastery of knowledge in their specific disciplines, UP graduates must possess breadth of mind, strength of character, and generosity of spirit, fostered by a firm grounding in both the arts and sciences, and such specialist courses as their programs may require. They must be prepared to inclusively engage with society and the world at large, mindful of their people's needs and capabilities, and keen to the challenges and opportunities of national development in this century of rapid global change. UP aims to achieve this through its General Education program, one that develops mind, body and spirit, which familiarize all its students with their culture and history and fosters a sense of shared citizenship, while equipping them with critical thinking, discernment and technical skills they will need to excel in their chosen professions.	UP Mindanao Vision An internationally competitive graduate, research, and public service university in Mindanao, leading in the study of arts, cultures, and sustainable biosystems. Mission The University of the Philippines Mindanao provides distinctive interdisciplinary education, research & development, and public engagement that are responsive and relevant to the diverse sectors in Mindanao and sustains Mindanao heritage and biodiversity through innovative systems and technologies. Department Vision The department's vision is to be an internationally recognized department working on Mathematics and Computing for human knowledge, research, innovation, and service. Department Mission The department provides world-class quality education, impactful research, and responsive extension services to diverse sectors in Mindanao and the rest of Southeast Asia through its academic programs in the sciences and engineering, and cultivates a passion for excellence, creative thinking, and commitment to the common good in the context of culturally diverse global community.	Program Learning Outcomes PLO 1 Apply appropriate knowledge, skills, and techniques in computer science, scientific computing and modeling, and computer science analysis and design in developing IT solutions for real-world problems, managing IT solution companies, and in doing multi-disciplinary research. PLO 2 Design and create relevant IT systems that meet the desired demands in the IT industry by integrating knowledge and skill in computer science and with that of other disciplines. PLO 3 Communicate effectively in oral and written communication to both technical and non-technical audiences. PLO 4 Demonstrate development of adaptive lifelong learning skills in learning new knowledge, skills, and techniques in computer science. PLO 5 Demonstrate leadership, professionalism, and the ability to work with members in a multi-disciplinary and multi-cultural team. PLO 6 Innovate computing solutions and technologies that will address environment-related problems and will improve the welfare of the Filipino people through operational efficiency, transaction transparency and transfer of technology. PLO 7 Develop platform-independent IT solutions and technologies using appropriate computing principles that are potentially patentable and commercially viable. PLO 8 Develop computing solutions and technologies to address socio-cultural needs and guided by professional, research, and ethical practices.
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COURSE INFORMATION

Course Number:	CMSC 3
Course Title:	Introduction to Web Design
Course Description:	Fundamentals of HTML, forms, webpage interaction creation and editing, HTML5, and effective design practices for a successful website design.
Course Prerequisite:	
Course Credit:	3 units
Offering period	First Semester

COURSE LEARNING OUTCOMES

After completing this course, you should be able to—

		<i>Program Learning Outcomes aligned to the CLOs</i>
CLO 1	Define the principles of Web Design;	PLO 1, 4
CLO 2	Demonstrate their webpage design capabilities from their understanding on the fundamentals of HTML;	PLO 1, 2
CLO 3	Demonstrate their implementation capabilities in the HTML/HTML5 language through proper coding of HTML Tags, links, forms, design, and interactions; and	PLO 1, 2, 7
CLO 4	Practice the best practices for creating successful website design.	PLO 2, 4, 8

COURSE OUTLINE

1. Introduction to Web Design
 - 1.1. History of the Internet
 - 1.2. Evolution of Websites
 - 1.3. Websites and Standards
 - Principles of web accessibility
 - Principle of Privacy & Security
 - Principles of Internationalization
2. Principles of Web Design
 - 2.1. Basic Principles in website development
 - 2.2. Planning your website design
 - 2.3. Website design and Layout
3. HTML Fundamentals
 - 3.1. Introduction to HTML
 - 3.1.1. What is HTML?
 - 3.1.2. HTML Versions
 - 3.1.3. HTML Documents
 - 3.1.4. HTML Tags
 - 3.1.5. HTML Page Structure
 - 3.2. Elements of HTML
 - 3.2.1. HTML Headings
 - 3.2.2. HTML Paragraphs
 - 3.2.3. HTML Attributes
 - 3.2.3.1. Tables
 - 3.2.3.2. Links
 - 3.2.3.3. Forms
 - 3.2.3.4. Styles



- 3.2.3.5. Formatting
- 3.2.3.6. Color
- 3.2.3.7. Quotations
- 3.2.3.8. Comments
- 4. Cascading Style Sheets (CSS)
 - 4.1. HTML Styling Using CSS
 - 4.1.1. Introduction to CSS
 - 4.1.2. Creating a Style Sheet
 - 4.1.3. CSS Properties
 - 4.1.4. CSS Styling
 - 4.1.4.1. Background
 - 4.1.4.2. Fonts
 - 4.1.4.3. Text Formats
 - 4.1.4.4. Color
 - 4.1.4.5. Borders
 - 4.1.4.6. Margins
 - 4.1.5. Lists and Tables
 - 4.1.6. CSS id and classes
 - 4.1.7. Advanced CSS

- 5. HTML 5
 - 5.1. Introduction to HTML 5
 - 5.2. New HTML 5 Elements
 - 5.3. Removed HTML Elements
 - 5.4. Semantic Elements
 - 5.5. HTML 5 Styling

MODE OF DELIVERY

The lecture class will be conducted in either face-to-face mode or asynchronous mode while the laboratory class will be in blended mode. The designated Learning Management System (LMS) will be used to deliver the lecture materials, which include videos, assignments/exercises, and assessments. Asynchronous meetings for live discussions/forums shall be conducted via Zoom, Google Meet, Microsoft Teams, or any relevant software.

For questions and other concerns, students may post them in the LMS discussion board or directly send a message to the instructor through the LMS messaging facility or the Google Space for the course. The instructor will address your queries promptly or during your class schedule.

COURSE MATERIALS

To facilitate the learning of this course, students need the following:

Material Name	Description
Course Guide	This document contains the course description, learning outcomes, study schedule, course requirements, and grading scheme. This must be read at the start of classes and, when necessary, revisited to keep the student on track with the progression of the course.
Study Guide	This is provided for each segment of topic, which may cover a week or two, to guide the students in their weekly activities and learning progress.
Lecture Presentations	Lecture slides will be given in PDF format (you may need to install a PDF reader for this). This will supplement the video lectures.
Video Lectures/ Tutorials	These are lectures or tutorials of the topics covered. Students need to watch these videos, which are created by your professor or educators from other universities or experts from the industry. Links are provided in the Study Guide posted in the LMS.
Computer	Any device e.g. desktop, laptop, tablet or smartphone, which has connection to the Internet to access the learning materials.
Integrated Development Environment (IDE)	Any IDE of their preferred programming language for the development of their image and video processing software project.



LEARNING ACTIVITIESStudy schedule in **Lecture and Laboratory**

Week #	Session #	Course Outcomes	Topics	Activities	References	Assessment
	Session 0	Ø Acquire an overview of the course syllabus and class policy	Course Syllabus	Open discussion	Course Syllabus	Oral recitation
Week 1-2	Lecture Session 1	Ø Describe the History and Evolution of Websites Ø Relate the different Standards of Websites	History of the Internet	Lecture Guided Open discussion	[1],[2],[3],[4],[5]	Oral recitation
Week 1-2	Lecture Session 2		Evolution of Websites	Guided Open discussion w/ Sample Websites	[1],[2],[3],[4],[5]	Written exercise



	Lecture Session 3	and identify the importance of having standards in Web Design	Websites and Standards	Guided Open discussion w/ Sample Basic Problems	[1],[2],[3],[4],[5]	Written exercise Take-home Exercise
	Laboratory Session 1-2		Evolution of Websites Websites and Standards	Guided Open Discussions Laboratory Exercises	[1],[2],[3],[4],[5], Lecture Notes	Written Exercises
Week 3	Lecture Session 4-5	Ø Describe different principles in website development Ø Relate the importance and impacts of the different principles in practicing website development	Basic Principles in Website Development	Guided Open discussion with Examples and Test Cases	[1],[2],[3],[4],[5]	Oral Recitation Take-home Exercise



	Laboratory Session 3	Ø Identify group mates from the class and form a project group for the project. Ø Discuss among group members their ideas and plan for the project		Grouping Consultation		Project Group Formed
Week 4-5	Lecture Session 6-9	§ Recognize the different stages of Website Design Planning § Recognize and Describe the different factors affecting the planning stage of Web design § Describe the best practices in Web Design Planning § Identify the elements of Website Design and Layouts	Planning your Website Design Website Design and Layout	Guided Open discussion with Examples and Test Cases	[1],[2],[3],[4],[5]	Written Exercise Oral Recitation



Week 4-5	Laboratory Session 4-5	§ Discuss among group members their ideas and plan for the project § Discuss among group members their ideas and plan for the Website Design and Layout	Planning your Website Design Website Design and Layout	Consultation Laboratory Exercise	[1],[2],[3],[4],[5], Lecture Notes	Written Exercises
	Lecture Session 10-12	§ Explain what HTML is and how it is different with other Programming Languages § Differentiate HTML Versions § Distinguish the different HTML Documents	What is HTML? HTML Versions HTML Documents	Lecture Guided Open discussion with Example HTML Documents	[1],[2],[3],[4],[5]	Oral Recitation Written Exercise



Week 6-7	Lecture Session 13	§ Demonstrate understanding of the different principles in web design § Summarize the different standards of web design § Explain how the different standards of web design impact Website design and programming	1 st Long Exam	Written Exam	[1],[2],[3],[4],[5]	Comprehensive Exam
	Laboratory Session 6-7	§ Practice how to create HTML Documents	HTML Documents	Laboratory Exercise on Creating HTML Documents	[1],[2],[3],[4],[5], Lecture Notes	Programming Exercises



Week 8-10	Lecture Session 14-19	§ Identify the different Elements of HTML § Distinguish the output of implementing the different elements of HTML in Web pages § Recognize the effects of the combination of the different HTML elements § Recognize the impact of the best practices in coding in HTML	Elements of HTML: § HTML Headings § HTML Paragraphs § HTML Attributes · Tables · Links · Forms · Styles · Formatting · Color · Quotations · Comments	Lecture, Program Tracing, Guided Open discussion with Sample HTML Documents Lecture, Program Tracing, Guided	[1],[2],[3],[4],[5]	Oral Recitation Written Exercise Take home exercise
	Laboratory Session 8	§ Demonstrate the implementation of the different elements of HTML § Demonstrate in coding the	Elements of HTML: § HTML Headings § HTML Paragraphs	Laboratory Exercise on HTML Headings and Paragraphs Project Update	[1],[2],[3],[4],[5]	



Week 8-10	Laboratory Session 9	output of the combinations of the different HTML elements in HTML Documents § Experiment on the effects of different formatting styles in Web pages such as the combination of different color and styles	Elements of HTML: § HTML Attributes · Tables · Links · Forms	Laboratory Exercise on HTML Tables, Links and Forms Project Update	Lecture Notes Project Consultation	Programming Exercises Updated Project Progress
	Laboratory Session 10		Elements of HTML: § HTML Attributes · Styles · Formatting · Color · Quotations · Comments	Laboratory Exercise on HTML Styles, Formatting, Color, Quotations, and Comments Project Update		



Week 11-13	Lecture Session 20-25	§ Identify the different Elements of CSS § Distinguish the output of implementing the different elements of CSS in designing Web pages § Recognize the effects of the combination of the different CSS Styling elements in Web page designs § Recognize the impact of the best practices in designing Web pages using CSS	§ HTML Styling Using CSS § Introduction to CSS § Creating a Style Sheet § CSS Properties § CSS Styling · Background · Fonts · Text Formats · Color · Borders · Margins § Lists and Tables § CSS id and classes § Advanced CSS	Lecture, Program Tracing, Guided Open discussion with Sample HTML Documents with CSS	[1],[2],[3],[4],[5]	Oral Recitation Written Exercise Take home exercise
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Week 11-13	Laboratory Session 11	§ Demonstrate the implementation of the different elements of CSS § Demonstrate in coding the styling output of the combinations of the different CSS elements in HTML Documents § Demonstrate the impact of Advanced CSS in Web page styling § Experiment on the effects of different formatting and styling in Web pages using CSS	§ HTML Styling Using CSS § Introduction to CSS § Creating a Style Sheet § CSS Properties § CSS Styling · Background · Fonts · Text Formats · Color · Borders · Margins	Laboratory Exercises on CSS Styling using different CSS elements. Project Update	[1],[2],[3],[4], [5] Lecture Notes Project Consultation	Programming Exercises Updated Project Progress
	Laboratory Session 12		§ HTML Styling Using CSS § Lists and Tables § CSS id and classes	Laboratory Exercises on CSS Styling Lists and Tables in HTML, and CSS coding using id and classes. Project Update		
	Laboratory Session 13		Advanced CSS	Laboratory Exercises on Advanced CSS Styling Project Update		



Week 14-16	Lecture Session 26	§ Demonstrate understanding of the proper implementation of HTML and CSS elements § Demonstrate understanding of the impact of the combination of different HTML and CSS elements in HTML Documents § Identify portions of the code that are implemented incorrectly or not as intended	2 nd Long Exam	Written Exam	[1],[2],[3],[4], [5] Lecture Notes	Limited F2F: Final Exam
	Lecture Session 27-30 Lecture Session 27-30	§ Identify the new HTML elements introduced for HTML 5 Documents § Identify the Removed HTML Elements for HTML 5 Documents § Understand the impact of both the new and removed HTML elements in Web page creation and designing § Differentiate HTML 5 implementation changes from basic HTML § Familiarize new styling mechanisms in HTML 5 implementation	§ HTML 5 · Introduction to HTML 5 · New HTML 5 Elements · Removed HTML Elements · Semantic Elements · HTML 5 Styling § HTML 5 · Introduction to HTML 5 · New HTML 5 Elements · Removed HTML Elements · Semantic Elements · HTML 5 Styling	Lecture, Program Tracing, Guided Open discussion with Sample HTML 5 Documents Lecture, Program Tracing, Guided Open discussion with Sample HTML 5 Documents	[1],[2],[3],[4], [5] [1],[2],[3],[4], [5]	Oral Recitation Take home exercise Oral Recitation Take home exercise



Week 14-16	Laboratory Session 14-15	§ Demonstrate the changes in HTML 5 from the basic HTML implementation § Experiment on the new semantic elements present only in HTML 5 § Experiment the effects on Web page design using HTML 5 styling		Laboratory Exercises on HTML 5 implementation	[1],[2],[3],[4], [5], Lecture Notes	Programming Exercise
	Laboratory Session 16		Project Update	Project Update		Updated Project Progress Project Submission



	Set by OUR	§ Demonstrate their capability to understand code based on the elements of HTML, HTML 5 and CSS § Demonstrate their ability to create aesthetic Web pages using HTML/CSS styling techniques § Explain the importance of adhering to the standards and best practices of Web page designing	Final Exam	Final Project Presentation	[1],[2],[3],[4], [5] Lecture Notes	Final Project Presentation
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REFERENCES

- [1]
- [2]
- [3]
- [4]
- [5]
- [6]



COURSE REQUIREMENTS

Students must comply the following course requirements:

Lecture		Laboratory	
Long Exams	60%	E-Portfolio	20 %
		Project (20% from Project Update Presentation)	30 %
		Laboratory Exercises	50 %
Report/Final Exam (Final Project Presentation)	20%		
Assignments/Quizzes	20%		
100%		100%	

The final grade for this course will be computed based on two components: **60%** from the Lecture component and **40%** from the Laboratory component.

Long Exam

The class will involve lectures to introduce the different concepts in webpage design. Lecture classes will also involve individual or collaborative practice exercises to verify student's understanding of the topics. Announced quizzes and lecture exams will be given during lecture sessions as well. Laboratory sessions, which are mostly collaborative in nature, will involve actual implementation of concepts discussed in lectures. On-the-spot-programming (OTSP) will also be administered on certain laboratory class sessions as laboratory exams. OTSP is expected to be accomplished on an individual basis. Students are encouraged to ask questions during classes and the instructor's consultation hours.

Quizzes could be announced or unannounced. No make-up quizzes will be given whether excused or unexcused. No make-up exams will be given (except to students whose absences are deemed excusable by the CSO). Complaints regarding the checking of exams/quizzes will only be entertained within one week of its release. Dates of examination and lectures stated above are subject to changes.

E-portfolio

All output produced by the student will be compiled.

Report

Each student will be assigned a topic. He/she needs to research and prepare a presentation about this topic. The grade in the report will be based in an oral report rubric.

GRADING SCALE

Final grades will be based on the following grading scheme:

96 and above	1.00	[75, 80)	2.25
[92, 96)	1.25	[70, 75)	2.50
[88, 92)	1.50	[65, 70)	2.75
[84, 88)	1.75	[60, 65)	3.00
[80, 84)	2.00	below 60	5.00



HOUSE RULES

Attendance:

There will be no grades of 4.0. A student who completes the requirements of the course will be given a passing mark. Failure to comply with the requirements of the course within the semester will mean a grade of 5.0.

Dropping. A student is officially dropped from the course if s/he successfully completes the dropping process. A student who stops attending classes and/or taking exams without officially dropping will be given a grade of (5.0) DRP. Note that the teacher has the right not to sign your dropping slip.

Missed Exams. A student who misses an examination should immediately inform the teacher personally. Depending on the merits of the situation, the marks on final exam will be used for that particular exam. If the student misses another examination, he/she will automatically be given a zero mark in that exam.

Academic Dishonesty. All forms of academic dishonesty will not be tolerated. Any student caught cheating will undergo investigation. If found guilty of cheating, s/he will receive a final grade of 5.0 for the course. The main reason for this is to highly discourage students from committing any form of academic dishonesty. Refer to Section 2 and Section 25 of the UP Diliman Faculty Manual.

Faculty in Charge:

Name: Gracie Anne R. Abot

Lecturer 1

Contact Info: 09658254269, grabot@up.edu.ph

Email me: If you have concerns that you wish to discuss with me, do not hesitate to email me. using your @up email address. In your email's subject, include the course number and then. your concern. For example, the subject may look like: CMSC 3 – Exam 1 Coverage.

CLASS POLICIES

Enrolling to this course means that you strongly adhere to the following class policies:



1. **Submission of Requirements and Deadlines.** Students must submit their work through the course Moodle LMS. Each course requirement has a deadline that the students should follow, otherwise their submission will be considered late, which will incur a deduction of 20% (round up) from the total score.
2. **Intellectual Honesty.** This course is supposed to give you knowledge and skills to better prepare you for your future career. However, this purpose will be defeated if you cheat on your works (e.g. copying of codes and submitting it as your own, etc.). Hence, all forms of academic dishonesty will not be tolerated. Any student(s) who are suspected or caught cheating will undergo investigation. If found guilty of cheating, s/he will receive a grade of 5.0 and will result in disciplinary action in accordance with University rules, policies, and procedures. Refer to Section 2 and Section 25 of the UP Diliman Faculty Manual.
3. **Technical difficulties.** Any problems (e.g. submission concerns, etc.) or other important concerns about the course should be raised to your instructor immediately. Do not hesitate to ask questions to your instructor concerning your course.
4. **Stay tuned.** Regularly check the course Moodle LMS for weekly updates (e.g. uploaded learning resources, requirements, assessments, etc.)

Further rules may be added later.

IMPORTANT INFORMATION

Academic Integrity

As a student of the University of the Philippines, you pledge to act ethically and uphold the value of honor and excellence.

You are encouraged to execute your tasks or activities truthfully and independently. You may work with others as permitted by the Instructor. With this, you understand that any suspected misconduct will be reported to the appropriate office for investigation and if established, will result in disciplinary action in accordance with University rules, policies, and procedures.

Copyright Notice

All materials in either printed or digital form are intended only for the use of the students enrolled in this course. These resources may contain works that are subject to copyright protection under RA 8293 or the “Intellectual Property Code of the Philippines”. Any reproduction and/or communication of any materials included in this course may be subject to copyright infringement and the copyright owners have the right to take legal action against such infringement.



Do not remove this notice.

Revision No	2	<i>Note: A revision of this course syllabus is considered when there are changes in CLOs, topics (more than 20% of the contents), and course requirements (with justification).</i>
Revised (Date)	August 29, 2023	
Revised by	Name of Faculty-in-charge	Signature
Evaluated by	Program coordinator	Signature
Endorsed by	Department Chair	Signature
Approved by	Dean	Signature

